

Total No. of Questions : 8]

SEAT No. :

PE2748

[Total No. of Pages : 2

[6583]-331

T.E. (Cyber Security) (Honour)
INFORMATION AND CYBER SECURITY
(2019 Pattern) (Semester - V) (310401)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of Scientific Calculator is permitted.

Q1) a) Explain RSA algorithm in detail with example. **[8]**

b) What is Chinese Remainder Theorem? Explain with example. **[9]**

OR

Q2) a) Explain in detail Diffie-Hellman with justification. **[8]**

b) Explain Elliptic Curve Arithmetic and Elliptic curve Cryptography. **[9]**

Q3) a) Explain the types of Cyber-Crimes. Explain Cybercrime scenario in India. **[9]**

b) What are challenges in Social engineering? Explain Cyber Stalking? **[9]**

OR

Q4) a) List the steps for risk identification and assessment in risk management for Information Security. **[9]**

b) What are the types of Laws? Differentiate between Law and Ethics. **[9]**

Q5) a) What protocols comprise SSL with neat diagram? What is the difference between SSL connection and SSL session? **[8]**

b) What services are provided by IPSec? Give difference between Transport mode and Tunnel Mode. **[9]**

OR

P.T.O.

- Q6)** a) What is the role of X.509 in cryptography? [8]
b) What are various types of firewall? Discuss limitations of firewall. [9]

- Q7)** a) What is a phishing attack explain with an example. What are the different types of phishing? [9]
b) What is computer worm and virus? How does computer virus spread? How to protect against computer virus and worms. [9]

OR

- Q8)** a) Write short note on Intrusion Detection System with Types. Limitations and challenges. [9]
b) What is DoS & DDoS Attacks? Explain with suitable examples. [9]

